



Fungal colonization and invasive fungal infections following allogeneic BMT using metronidazole, ciprofloxacin and fluconazole or ciprofloxacin and fluconazole as intestinal decontamination

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Summary:

Invasive fungal infections (IFI) are increasingly diagnosed in patients undergoing allogeneic BMT. We have previously shown that the addition of metronidazole to ciprofloxacin for gastrointestinal bacterial decontamination significantly reduces the incidence of grades II–IV aGVHD by reduction of the anaerobic intestinal bacterial flora. Here, we found that the combined use of ciprofloxacin, metronidazole and fluconazole as antifungal prophylaxis increased intestinal yeast colonization when compared to ciprofloxacin and fluconazole alone ($P < 0.01$). Based on the EORTC criteria, a total of 18 out of 134 study patients developed IFI: seven of 68 (10%) patients who received metronidazole compared to 11 of the 66 (17%) patients decontaminated without metronidazole developed IFI (log-rank $P = 0.36$). Lethal IFI occurred in two of seven patients receiving metronidazole and in four of 11 patients without anaerobic decontamination. In conclusion, bacterial intestinal decontamination using metronidazole as an antibiotic with activity against most anaerobic intestinal bacteria significantly increases the intestinal yeast burden without influencing the incidence of IFI in patients undergoing allogeneic BMT. *Bone Marrow Transplantation* (2000) 26, 993–997.

Keywords: invasive fungal infection; BMT; metronidazole; total gut decontamination; aGVHD

Alloreactive T cell activation follows differences between recipient and the donor gene products of MHC or polymorph tissue antigens outside the MHC. There are well established additional factors influencing the development of aGVHD such as type of immune prophylaxis, patient and donor age, underlying disease, and state of alloimmunization of female donors donating marrow for male recipients.^{3–6} As described in early studies, germ-free or completely decontaminated rodents show a decrease in aGVHD even in recipients of MHC mismatched transplants.^{7,8} The underlying mechanisms of modulating aGVHD by gastrointestinal decontamination remains unclear. However, it has been postulated that reduction in aGVHD increases immune recovery not only by virtue of reduced immunosuppressive therapy but also because of a reduced immunosuppressive effect of aGVHD itself. Several studies designed to realize total gut decontamination including anaerobic microflora were initiated in this context. In a recent single center open label study, we have shown that antimicrobial chemotherapy targeted to intestinal anaerobic bacteria in marrow transplant recipients significantly reduces the severity of aGVHD. This supports the theory that intestinal anaerobic bacterial microflora play an essential role in the pathogenesis of aGVHD after BMT.⁹ After total gut decontamination using metronidazole to reduce intestinal anaerobic microflora, yeast overgrowth is evident.¹⁰ Here, we report our experience on the impact of fungal overgrowth on invasive fungal infections (IFI) in these study patients.

Patients and methods

One hundred and thirty-four patients (15–57 years of age) with hematologic malignancies were enrolled in this study. All patients underwent allogeneic BMT between September 1993 and August 1995. Patients had either a genotypically HLA-identical donor, a partially HLA-matched extended family donor, or a volunteer matched unrelated donor according to the German consensus criteria.¹¹ The study protocol had been approved by the Ethics Committee at the Medical Faculty of the University of Essen (Essen, Germany).

The study was performed as a single-center open-label prospective randomized trial, which compared two different regimens of intestinal bacterial decontamination in order to investigate the incidence of aGVHD. Patients randomly

Invasive fungal infections (IFI) are increasingly diagnosed in immunocompromised patients after allogeneic stem cell transplantation (SCT). Once a diagnosis of IFI has been made, mortality is high. Prevention of aGVHD is an important factor determining transplant-related morbidity and mortality caused by IFI.^{1,2} This is not only due to a decreased administration of immunosuppressive drugs but also to the immunosuppressive effect of aGVHD by itself.

Acute GVHD results from an inflammatory process of recipient tissues that is induced by donor T cell clones.

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assigned to the combined intestinal bacterial decontamination medication were treated with oral metronidazole at 400 mg three times daily and ciprofloxacin 750 mg twice daily p.o. (study arm A). Patients on single bacterial decontamination received oral ciprofloxacin 750 mg twice daily p.o. (study arm B). As fungal prophylaxis, all patients received oral fluconazole 200 mg once daily p.o. The primary end point of the study was designed to evaluate the incidence of a maximal clinical grade of aGVHD greater than grade I in the two different arms as well as a quantitative comparison of intestinal anaerobic growth suppression. Treatment according to the study arm was initiated on day -14 where day 0 designates the infusion of bone marrow. The decontamination regimen was given until day 35 post transplant, diagnosis of acute GVHD, or death, whichever came first. These data have been published.⁹ A secondary analysis concerning intestinal colonization with yeast strains and development of invasive mycosis was realized. Follow-up was continued until the death of the patient or at least over a period of 6 years after BMT. Patient characteristics are summarized in Table 1.

Supportive care

All patients were nursed in reverse isolation rooms equipped with high efficiency particle air filtration (HEPA)

systems. These conditions were introduced on day -7 and usually maintained until day +35 post transplant. Aseptic techniques were used during patient contact throughout this time period. Blood component substitution, oral and parenteral nutrition, and treatment of suspected or documented bacterial or mycotic infections followed the published guidelines, as did strategy for monitoring herpes simplex or cytomegalovirus replication or infection.^{9,12}

Microbiologic analysis

Fecal samples of the study patients were monitored twice weekly for growth of aerobic and anaerobic bacteria as well as fungi, using modified microbiologic culture techniques as previously published.^{10,12} Quantitative bacterial cultures were taken using a conventional plate counting technique. Standardized dilutions of stool samples were cultured on several aerobic and anaerobic as well as fungal culture agars (CLED agar, *Clostridium difficile* selective agar with D-cyclo serine and cefoxitin, Mannitol salt agar, Rose Bengal Chloramphenicol agar, Yeast extract Cystine blood agar with gentamicin and nalidixic acid, and Drigalski agar). Bacteria were categorized according to culture growth conditions as either aerobic or anaerobic. Single colonies were identified by conventional biochemical tests. Quantification of bacterial culture growth was expressed as

Table 1 Patient characteristics of the study population

	Ciprofloxacin + fluconazole + metronidazole	Ciprofloxacin + fluconazole	P value
No. of patients	68	66	
Median age (years) (range)			
Patients	36 (17–56)	37 (15–57)	
Donors	37 (14–64)	38 (22–59)	NS
Donor type			
HLA-identical sibling donor	40 (59)	40 (61)	
Partially HLA-matched family donor	8 (12)	11 (17)	NS
Matched unrelated donor	20 (29)	15 (23)	
Ratio sibling donor to alternative donor	0.41	0.40	
Diagnosis			
Acute lymphoblastic leukemia	2 (3)	6 (9)	
Acute myelogenous leukemia	20 (29)	16 (24)	
Chronic myelogenous leukemia	39 (57)	39 (59)	
Malignant lymphoma	1 (1)	1 (2)	NS
Multiple myeloma	4 (6)	1 (2)	
Myelodysplasia	2 (3)	3 (5)	
Disease stages			
Early stages	35 (51)	41 (62)	
Advanced stages	33 (49)	25 (38)	NS
GVHD prophylaxis			
Short methotrexate + cyclosporine	49 (72)	48 (73)	
Cyclosporine	19 (28)	18 (27)	NS
aGVHD			
Grade 0	16 (24)	14 (21)	
Grade I	35 (51)	19 (29)	
Grade II	5 (7)	12 (18)	<0.04
Grade III	3 (4)	6 (9)	
Grade IV	9 (13)	15 (23)	
Cumulative probability of grade II–IV aGVHD	25 ± 5%	52 ± 6%	<0.002
Myeloablative regimen			
TBI + cyclophosphamide	52 (76)	55 (63)	
Chemotherapeutic regimen	16 (24)	11 (17)	NS
Cytomegalovirus-negative donor–recipient combination	24 (35)	17 (26)	NS

NS = non-significant; TBI = total body irradiation.

the log₁₀ of colony-forming units with a detection threshold below 10³ c.f.u. per gram of sample. For the purpose of analysis on bacterial culture growth suppression, cultures with no c.f.u. growth were calculated as 10⁰ c.f.u. In cases of diarrhea, fecal samples were additionally screened for enterotoxins, enteropathic viruses, and parasites.

Statistical analysis

For comparison of timely repeated continuous measures of stool colonization between the two groups, repeated measures analysis of variance was applied. Differences between frequencies were compared using the two-tailed Fisher's exact test (2 × 2 frequency tables) or by the Mantel–Haenszel chi-square test (2 × *n* frequency tables). For the cumulative probabilities of grades II to IV aGVHD and IFI, Kaplan–Meier product-limit estimates were calculated. Comparisons between the time to event-curves were made by log-rank statistics as previously published.⁹

Results

Intestinal fungal colonization

Eight hundred and sixty-six fecal samples were evaluable for bacterial culture growth during the post-transplant course. For yeast growth, a total of 916 samples was evaluable. The proportion of fecal samples with no detectable anaerobic bacterial growth was two-fold higher (236 of 446 (53%)) in patients receiving metronidazole, ciprofloxacin and fluconazole compared to transplant recipients receiving ciprofloxacin and fluconazole (96 of 420 (23%)). The proportion of fecal samples with no detectable yeast growth was 1.3-fold higher in patients without suppression of anaerobic bacteria (334 of 441 (76%)) compared to patients receiving metronidazole (293 of 475 (62%)). Using repeated measures analysis of variance to compare the fecal germ concentrations between the two groups, we found a significant decrease for anaerobic bacterial growth in the post-transplant course (*P* < 0.001). Similarly, an increase in yeast colonization for patients receiving metronidazole was demonstrable (*P* < 0.01).

Invasive fungal infection (IFI)

Diagnosis of IFI was based on published EORTC criteria.¹³ A total of 18 out of 134 (13%) study patients developed IFI. There was no difference in the incidence of IFI according to the different decontamination regimens using a Kaplan–Meier analysis: seven of 68 (12 ± 3%) patients receiving metronidazole compared to 11 of 66 (17 ± 6%) patients decontaminated without metronidazole developed IFI. The cumulative estimates of IFI showed no difference between the two patient subsets (log-rank test *P* = 0.36).

Study arm A included six of seven patients with proven IFI and one patient with probable IFI. In study arm B, eight of 11 patients had proven, two of 11 patients had possible, and one of 11 patients had probable IFI (Figure 1).

Tables 2 and 3 illustrate the characteristics of patients with IFI. *Aspergillus fumigatus* was the leading isolate in

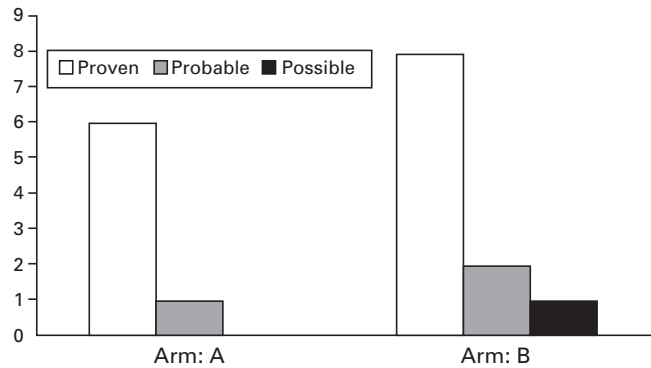


Figure 1 Number of patients with diagnosed invasive fungal infections according to the EORTC criteria: proven, probable, or possible invasive fungal infection. Arm A: gut decontamination including metronidazole, ciprofloxacin, and fluconazole. Arm B: gut decontamination including ciprofloxacin and fluconazole.

both groups. Emergence of IFI caused by *A. fumigatus* and *Candida* species was equally distributed in the two arms.

Out of the 18 documented cases of IFI, eight patients (44% of all diagnosed patients with IFI and 6% of the entire patient population) died from fungal infection as the primary cause of death. In study arm A, we noted one death due to *Aspergillus* encephalitis and one case of candidemia caused by *C. krusei*. In patients receiving no anaerobic decontamination, four cases of invasive aspergillosis were detected. Two patients were diagnosed with invasive pneumonia by *A. fumigatus*, one patient with septicemia by *A. fumigatus*, and one patient with pneumonia caused by an unclassified *Aspergillus* species. Further, we observed two candidemias caused by *C. krusei* and by an unclassified *Candida* species. Thus, total bacterial intestinal decontamination using metronidazole and ciprofloxacin was not associated with an increase of primary lethal IFI. Introduction of metronidazole to a standard bacterial intestinal decontamination medication also showed a trend to decreased primary mortality by invasive mycosis (*P* = 0.13).

In each arm, one patient was successfully treated by amphotericin B. One patient (1/68) had a proven relapsing urinary tract infection by *C. glabrata* (study arm A) and the other (1/66) was cured from a probable aspergillus pneumonia (study arm B).

Discussion

As previously described, the effects of broad-spectrum antibiotics change the gastrointestinal yeast flora of humans. The present results confirm prior findings obtained from mouse models of gastrointestinal colonization by *Candida albicans* in response to broad-spectrum antibiotics. Antibiotics with anaerobic activity or high intestinal concentrations particularly cause a higher and more sustained increase in gastrointestinal yeast colonization than do antibiotics with a poor anaerobic activity or a low gastrointestinal concentration.¹²

The introduction of oral metronidazole to a standard gut decontamination regimen in patients undergoing BMT increases the decontamination efficacy by specific antibac-

Table 2 Patient characteristics with IFI receiving ciprofloxacin, metronidazole and fluconazole as gut decontamination

No.	Disease	Fungal strain	EORTC diagnostic criteria	Fungal-related infection	Primary cause of death	Time of onset IFI diagnosis (days after BMT)
1	AML	<i>A. fumigatus</i>	probable	Pneumonia	GVHD	76
2	CML	<i>A. fumigatus</i>	proven	Pneumonia	CMV pneumonia	133
3	CML	<i>A. fumigatus</i>	proven	Encephalitis	IFI	165
4	MM	<i>C. albicans</i>	proven	Candidemia	Toxicity	44
5	CML	<i>C. glabrata</i>	proven	Urinary infection	Alive	14
6	CML	<i>C. krusei</i>	proven	Candidemia	IFI	14
7	CML	<i>C. tropicalis</i>	proven	Candidemia	Toxicity	39

IFI = invasive fungal infection; AML = acute myelogenous leukemia; CML = chronic myeloid leukemia; MM = multiple myeloma; A = aspergillus; C = Candida; GVHD = graft-versus-host disease.

Table 3 Patient characteristics with IFI receiving ciprofloxacin and fluconazole as gut decontamination

No.	Disease	Fungal strain	EORTC diagnostic criteria	Fungal-related infection	Primary cause of death	Time of onset IFI diagnosis (days after BMT)
8	AML	<i>A. fumigatus</i>	proven	Pneumonia	IFI	3
9	CML	<i>A. fumigatus</i>	proven	Pneumonia	IFI	121
10	CML	<i>A. fumigatus</i>	proven	<i>Aspergillus</i> septicemia	IFI	179
11	CML	<i>Aspergillus</i> spp.	possible	Pneumonia	Bacterial sepsis	171
12	ALL	<i>Aspergillus</i> spp.	probable	Pneumonia	IFI	315
13	CML	<i>Aspergillus</i> spp.	proven	Pneumonia	GVHD	51
14	AML	<i>Aspergillus</i> spp.	probable	Pneumonia	alive	28
15	ALL		proven	Invasive esophagitis	GVHD	63
16	CML	<i>C. krusei</i>	proven	Pneumonia	IFI	635
17	AML	<i>Candida</i> spp.	proven	Superinfected granuloma in sarcoidosis	Toxicity	244
18	CML	<i>Candida</i> spp.	proven	Candidemia	IFI	193

IFI = invasive fungal infection; AML = acute myelogenous leukemia; CML = chronic myeloid leukemia; ALL = acute lymphoid leukemia; A = aspergillus; C = candida; GVHD = graft-versus-host disease.

terial activity against most anaerobic strains compared with ciprofloxacin as single agent. We have previously demonstrated a definite and significant reduction of intestinal anaerobic bacterial flora.⁹

As anticipated, the present analysis confirmed a significant increase in gastrointestinal colonization with yeast strains in patients receiving metronidazole. However, this had no clinical impact on IFI, because the incidence of overall IFI as well as IFI as the primary cause of death were equally distributed in the two study arms. Despite the increased yeast colonization of patients receiving metronidazole, the number of yeast strains causing IFI was identical in both study arms. Therefore, the increased yeast colonization of patients receiving total bacterial intestinal decontamination did not seem to play a role in acquiring IFI in this patient population. The present analysis is sufficiently powered to prove an equivalent incidence of IFI assuming a maximum difference of 24%. Since we observed IFI in 10% of patients receiving metronidazole and in 17% of patients decontaminated without metronidazole, it appears justified to conclude that the incidence of IFI is similar in both patient subsets.

The integrity of the mucosal surface in patients with aGVHD of the gut is altered. With a decontamination regi-

men containing metronidazole the incidence of GVHD especially of the gut and the liver is significantly reduced.⁹ Therefore, there should be fewer gastrointestinal lesions facilitating yeast dissemination and in consequence IFI by *Candida* species.

We observed more *Aspergillus* infections in patients using ciprofloxacin as a single agent ($n = 7$) compared with patients receiving total gut decontamination ($n = 3$), but this difference did not reach significance ($P = 0.17$). Influence of the immunosuppressive effect of aGVHD and its inherent therapy on *Aspergillus* infections is well documented.^{1,2} In considering the reduced clinical grades of severity of aGVHD and the resulting lower intensity of immunosuppressive treatment, it is tempting to speculate that the use of metronidazole lowered the risk of opportunistic invasive *Aspergillus* and also *Candida* infections.

In summary, in patients undergoing allogeneic BMT, intestinal bacterial decontamination including metronidazole is associated with a significantly higher intestinal yeast burden without increasing the incidence of invasive fungal infection. This observation may be due to the metronidazole-mediated prophylaxis of aGVHD which may result in better host defense against invasive fungal infections.

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